

## SOME METHODS FOR ANALYSING ASSESSMENTS OF WRITTEN EXAMINATIONS

**Ross S. Sparks :** Department of Mathematical Statistics, University of  
Cape Town, Private Bag, Rondebosch 7700,  
Republic of South Africa.<sup>1</sup>

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residuals.

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**Summary :** Statistical techniques will be proposed to check:

- (a) whether raters are sequentially consistent as they progress with assessing candidates' performances on the same task;
- (b) whether assessment strategies of the raters differ.

An application will be used to illustrate these techniques.

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### 1. Introduction

The use of written examinations to assess competence is widespread, especially at educational institutions. The consequences of these assessments to the future opportunities open to the individual candidates do not need to be emphasised. Clearly, it is important to ensure that the assessment procedures are as fair as possible, and in particular, that they are free from inconsistencies and bias. This paper outlines a number of

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<sup>1</sup>Now at: Department of Mathematics, University of Wollongong,  
Wollongong, NSW 2500, Australia.

statistical techniques for detecting the inconsistencies that can occur when raters assess and quantify the achievement of candidates in written examinations.

The subjective nature of assessment is well known (Cronbach *et al.* 1972); different raters have different standards and also tend to differ in what they regard as important, that is they have different rating strategies. In addition, individual raters are subject to sequential inconsistencies, firstly as a result of factors such as mood and fatigue, and secondly because they tend to adjust their rating strategy over time, perhaps to meet other objectives, such as a desired pass rate. It might be possible to eliminate sequential inconsistencies if each paper were assessed more than once. However in practice assessments are often made under time constraints and, if the number of candidates is large, it is not unusual for the majority of papers to be assessed only once. In such circumstances the only way to deal with potential inconsistencies is to detect and correct them while assessment is in progress. This paper discusses procedures to automatically monitor assessments sequentially, and thereby alert raters who have changed their assessment strategy or whose strategy differs from the desired norm.

Substantial literature on the reliability of assessment procedures exist (e.g. Cronbach *et al.* 1972, Conger 1983, Boodoo & Garlinghouse 1983, Rae 1984, De Shields *et al.* 1984, Frary & Zimmerman 1984). Most of these analyses related to controlled experiments. This paper proposes techniques for comparing raters when each assesses mutually exclusive student performances of the same task. This is accomplished by making use of supplementary information, such as past performances in related tasks. In particular attention is given to techniques that would be useful in detecting:

- (i) differences in the assessment strategy of different raters;

- (ii) rater errors, such as errors resulting from incorrect addition of scores, unmarked answers, or inconsistent rating of the answers to particular questions;
- (iii) changes in rater strategy during the course of assessment, that is situations where a given level of achievement is assigned a higher/lower score (on average) in a later answer paper than it was in an earlier one.

The efforts needed to apply the techniques described in this paper is small compared to that required for the preparation and assessment of written examinations. Thus, considering the importance of making accurate assessments, there can be no excuse for not making use of these (or some other) techniques to improve the quality of assessments. Finally, the methods described here were designed to help raters improve their consistency. They are likely to be counter-productive if they are used to single out and chastise them.

## 2. Supplementary information — can it be useful?

Can raters check their assessments of candidates if they have prior knowledge of the candidate's previous achievements in related tasks? Through the use of a model, this section illustrates that this is possible.

The  $i$ -th candidate's "true" score (denoted  $w_i$ ) in particular task is a random variable with mean  $\mu_{w_i}$  and variance  $\tau_{ew_i}^2$ . Similarly the  $i$ -th candidate's "true" score (denoted  $z_i$ ) for the related task (i.e. supplementary information) is random with mean  $\mu_{z_i}$  and variance  $\tau_{ez_i}^2$ .

It will be assumed that  $\tau_{ew_i}^2 = \tau_{ew}^2$ ;  $\tau_{ez_i}^2 = \tau_{ez}^2$  and

$\begin{pmatrix} w_i \\ z_i \end{pmatrix}; i=1, \dots, n$  are independently distributed as  $n \left[ \begin{pmatrix} \mu_{w_i} \\ \mu_{z_i} \end{pmatrix}; \begin{pmatrix} \tau_{ew}^2 & 0 \\ 0 & \tau_{ez}^2 \end{pmatrix} \right];$

for all  $i=1, \dots, n$  i.e.  $w_i - \mu_{w_i}$  and  $z_i - \mu_{z_i}$  are normal and independent for

all  $i$ . Suppose that the between mean variations and covariations for all candidates in the population are defined by the distribution

$$\begin{pmatrix} \mu_{w_i} \\ \mu_{z_i} \end{pmatrix} \sim n \left[ \begin{pmatrix} \mu_w \\ \mu_z \end{pmatrix}; \begin{pmatrix} \tau_{Bw}^2 & \tau_{wz} \\ \tau_{wz} & \tau_{Bz}^2 \end{pmatrix} \right]$$

for all  $i$ , then

$$\begin{pmatrix} w_i \\ z_i \end{pmatrix} \sim n \left[ \begin{pmatrix} \mu_w \\ \mu_z \end{pmatrix}; \begin{pmatrix} \tau_w^2 & \tau_{wz} \\ \tau_{wz} & \tau_z^2 \end{pmatrix} \right]$$

where  $\tau_w^2 = \tau_{Bw}^2 + \tau_{ew}^2$  and  $\tau_z^2 = \tau_{ez}^2 + \tau_{Bz}^2$ .

The assessed scores (denoted  $y_i$  and  $x_i$  for the task and related task respectively) are also assumed to be randomly distributed about the "true" scores, that is

$$y_i = w_i + e_{y_i} \text{ and } x_i = z_i + e_{x_i}$$

where

$$\begin{pmatrix} e_{y_i} \\ e_{x_i} \end{pmatrix} \text{ are iid } n \left[ \begin{pmatrix} 0 \\ 0 \end{pmatrix}; \begin{pmatrix} \sigma_y^2 & 0 \\ 0 & \sigma_x^2 \end{pmatrix} \right].$$

In other words

$$\begin{pmatrix} y_i \\ x_i \end{pmatrix} \sim n \left( \begin{pmatrix} \mu_w \\ \mu_z \end{pmatrix}; \begin{pmatrix} \sigma_y^2 + \tau_w^2 & \tau_{wz} \\ \tau_{wz} & \sigma_z^2 + \tau_z^2 \end{pmatrix} \right).$$

The related scores are assumed known. Hence, of interest is the variation of  $y_i$  given the information  $x_i$ . This is best reflected by the regression model formed by conditioning on the score of the supplementary variable being given, i.e.

$$(1) \quad y_i = \beta_0 + \beta_1 x_i + e_i$$

where

$$\beta_0 = \mu_w - \beta_1 \mu_z; \quad \beta_1 = \tau_{wz} / (\tau_z^2 + \sigma_x^2);$$

$$e_1, \dots, e_n \text{ are iid } n(0, \sigma^2) \text{ and } \sigma^2 = \sigma_y^2 + \sigma_w^2 - \tau_{wz}^2 / (\sigma_x^2 + \tau_z^2).$$

Although the observed values  $y_1, \dots, y_n$ , may not be sampled independently, note that it will be assumed that the corresponding error components  $e_1, \dots, e_n$  are independent.

The closer the correlation  $\rho^2 = \tau_{wz}^2 / (\tau_z^2 + \sigma_y^2)(\tau_z^2 + \tau_x^2)$  is to one, the more useful  $x$  is in helping to detect errors in  $y$ .

On observing  $y$ , if  $e = y - (\beta_0 + \beta_1 x)$  is found to be large positive (negative), then this could be due to either:

- $e_w$  being large positive (negative) - meaning an unusually good (poor) performance by the candidate,

- $e_y$  being large positive (negative) - meaning a poor assessment made by a rater, resulting in an inflation (deflation) of the candidate's score for the task,
- $e_z$  being large negative (positive) - meaning an unusually poor (good) performance by the candidate in the related score, or
- $e_x$  being large negative (positive) - meaning a rater error could have deflated (inflated) the candidate's related score.

If the related score is an average of several assessed scores made on independent occasions, then  $e_z$  and  $e_x$  may be expected to be small in absolute value. In such situations, errors in  $e$  can be attributed mostly to either  $e_w$  and  $e_y$ .

Thus, a residual analysis (Draper & Smith 1981, chapter 3, Cook & Weisberg 1982, chapter 2) in regression model (1) could be useful in detecting poor assessments.

### 3. How can the assessment be checked for sequential inconsistencies?

If the related task  $x_1$  is an average of previously assessed identical tasks, then the sequential analysis of  $y_i - x_i$  for each rater  $i$  would be useful in detecting errors in assessments. However, if the task rated is more (less) difficult than the related task then the regression model (1) may be more appropriate for checking the assessments sequentially. An explanation will now follow.

In model (1),  $\beta_0$  and  $\beta_1$  are usually unknown and need to be estimated. These can be estimated using a starting number of assessed  $y$  scores and then updated with each new assessment, i.e. assume there are  $n$  candidates and the  $i$ -th candidate's related score is  $x_i$ , then the estimates of  $\beta_0$  and  $\beta_1$ , based on the first  $i$  candidates, are

$$(2) \quad \hat{\beta}_1(i) = \frac{\sum_{j=1}^i (x_j - \bar{x}_i)(y_j - \bar{y}_i)}{\sum_{j=1}^i (x_j - \bar{x}_i)^2}$$

and

$$(3) \quad \hat{\beta}_0(i) = \bar{y}_i - \bar{x}_i \hat{\beta}_1(i)$$

where  $\bar{a}_i = \sum_{j=1}^i a_j / i$ ;  $a = x$  or  $y$ , for  $i = 2, \dots, n$ . The quantity

$$(4) \quad r_{i+1} = \frac{y_{i+1} - \hat{\beta}_0(i) - x_{i+1} \hat{\beta}_1(i)}{\sqrt{1 + \frac{1}{i} + (x_{i+1} - \bar{x}_i)^2 / \sum_{j=1}^i (x_j - \bar{x}_i)^2}}$$

$i = 2, \dots, n - 1$  is called the recursive residual for the  $(i+1)$ -th assessment (see for example Galpin and Hawkins (1984)). The recursive residuals are useful for detecting systematic errors and outlying scores (e.g. unusually low or high scores for  $y$ ). If the observations are uncorrelated then, unlike the ordinary residual, the recursive residuals are independently distributed. Recursive residuals have the following interpretations:

- If the  $r_i$ -values for  $i=3, \dots, n$  are almost all negative (positive) then the rater is assigning lower (higher) scores on average with each new assessment;

- A large negative (positive)  $r_i$ -value would indicate either a rating error or an unusually poor (good) achievement by the  $i$ -th rated candidate;
- If the  $r_i$ -values are at first random about zero but later follow some trend (i.e. nonrandomness), then this would indicate a change in rating strategy;
- If the  $r_i$ -values are random about zero but increase (decrease) in absolute magnitude with each new assessment then the ratings are progressively made with less (more) accuracy.

The recursive residuals as calculated in equation (4) can be unreliable initially (i.e.  $i$  small), in particular, if errors are made in the first few assessments. These errors can be found by calculating the recursive residuals by reversing the order of the observation and then proceeding as in equation (4).

To illustrate individual case discrepancies a Shewhart chart, which is a straight index plot of the recursive residuals, is useful.

Sequential inconsistencies are best displayed by CUSUM (Cumulative Sum) recursive residuals plots (Brown *et al.* 1975, Galpin & Hawkins 1984). These will be discussed in the next section.

The Shewhart charts and CUSUM plots can be used to detect differences between the assessment strategies of several raters. For example, assume that two raters assigned scores  $y_1, \dots, y_{n_1}$  and  $y_{n_1+n_2}$  respectively, to  $n_1+n_2$  candidates. Define the related supplementary scores as  $x_1, \dots, x_{n_1}$  and  $x_{n_1+1}, \dots, x_{n_1+n_2}$  respectively. Treat these as a sequence of assessments made by a single rater and examine the recursive residuals

$r_1, \dots, r_{n_1+n_2}$ . If the  $r_i$ -values are random about zero for  $i = 1, \dots, n_1$ , but thereafter almost all  $r_i$ 's are negative or almost all  $r_i$ 's are positive then the two raters differ in assessment either on average or in terms of their scale. If however, the values  $r_1, \dots, r_n$  are random about zero but the last  $n_2$  recursive residuals are on average larger in absolute magnitude than the first  $n_1$  recursive residuals, then the second rater is more erratic than the first in his/her assessment.

#### 4. Cusum plots and their interpretation

If raters assign slightly lower scores, on average, as they progress with the assessment task then case diagnostic techniques, such as the Shewhart charts will be slow in detecting this trend. However if these effects are accumulated in some way then the detection of such trends may be more rapid. The CUSUM recursive residuals accumulate such effects using a scale invariant measure:

$$\text{CUSUM}_i = \sum_{j=3}^i r_j / s$$

where

$$(n-2)s^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2,$$

$$\hat{\beta}_1 = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) / \sum_{i=1}^n (x_i - \bar{x})^2,$$

$$\hat{\beta}_0 = \bar{y} - \bar{x} \hat{\beta}_1 \text{ and } \bar{a} = \sum_{i=1}^n a_i / n.$$

If  $e \sim n(0, \sigma^2)$  as in model (1) then  $CUSUM_i$  plotted against the index  $i$ , should show a random walk, with all  $CUSUM_i$  values lying within a parabolic envelope about the line  $CUSUM_i = 0$ . The following interpretations can be assigned to plots that deviate from a random walk:

- (i) If the  $CUSUM_i$  values show a downward (upward) trend then this will correspond to the rater assigning lower (higher) scores on average for each new assessment. This downward (upward) trend is caused by a series of negative (positive) recursive residuals (see the second half of figure 1(2));
- (ii) If the plotted  $CUSUM_i$  point shows a sudden jump away from earlier points, and thereafter the  $CUSUM_j$ ;  $j = i + 1, \dots, n$  values gradually drift back towards zero, then the  $i$ -th assessed score is unusual (see the first half of figure 1);
- (iii) If the  $CUSUM_0$  point is situated far from zero and the points  $CUSUM_j$ ;  $j = 1, \dots, n$  gradually drift towards zero then an assessment error may be made with the third candidate's performance but no serious errors thereafter (see the first half of figure 2).

The  $CUSUM_i$  plot is useful for detecting changes in strategy affecting  $\beta_0$  or  $\beta_1$  in model (1) but not in detecting changes in  $\sigma^2$ .

## 5. Detection of heteroscedastic recursive residuals

Two measures for detecting changes in residual variances  $\sigma^2$  will be

considered. Both of these are applicable only if the assessment strategy does not change in linear trend (i.e. if the regression coefficients  $\beta_0$  and  $\beta_1$  in model (1) remain constant) as assessment progresses. If the  $CUSUM_i$  values follow a random walk for all  $i$ , then

$$CUSS_i = \sum_{j=3}^i r_j^2 / \sum_{j=3}^i r_j^2, \quad i = 3, \dots, n$$

and

$$CUSAB_i = \sum_{j=3}^i \{(|r_j|/s)^{\frac{1}{2}} - 0,82218\} / 0,34914, \quad i = 3, \dots, n$$

(Hawkins 1981) can be used to detect heteroscedastic recursive residuals. If  $e_i$  is as defined in equation (1), then for large  $n$   $CUSAB_i$  is approximately normal in distribution with mean zero and variance  $i-2$  (Hawkins 1981) and  $(n-i)CUSS_i / (i-2)(1-CUSS_i)$  is distributed as a central  $F$ -distribution with  $n-i$  and  $i-2$  degrees of freedom. Under these assumptions,  $E(CUSS_i) = (i-2)/(n-2)$  for  $i = 3, \dots, n$ .

Assume that the sequence  $r_1, \dots, r_n$  randomly takes on negative or positive values, however  $|r_i|/s$  for  $i = m, \dots, n$  are on average unusually larger than the first  $m-1$   $|r_i|/s$  values, i.e.  $(|r_i|/s)^{\frac{1}{2}} - 0,82218$ 's are almost all greater than zero for  $i = m, \dots, n$ . This indicates that the rater is suddenly more erratic after the  $(m-1)$ -th assessment. In such situations, assuming no changes in regression parameters, the variance of the residual for the regression fit in equation (1) has increased, i.e. the recursive residuals  $r_j$  varying more for  $j \geq m$ . Assume that the plot of  $CUSUM_i$  versus  $i$  showed no trends, then figures 3 and 4 illustrate situations where scale changes are evident.

The interpretation of the  $CUSAB_i$  plot is similar to that of the

CUSUM<sub>*i*</sub> plot, except that the former plot is used to detect heteroscedastic residuals.

If the recursive residuals are homoscedastic then the plot of CUSS<sub>*i*</sub> against *i* should be approximately linear with slope equal to  $\frac{1}{n-2}$  and intercept  $\frac{-2}{n-2}$ . Figure 5a and figure 5b indicate the CUSS<sub>*i*</sub> plotted against *i* for situations when the recursive residuals increase and decrease in variance for increasing *i* respectively.

## 6. Tests for strategy differences amongst two or more raters

Assume that the strategy for the *i*-th rater is given by

$$(6) \quad y_{ji} = \beta_0^{(i)} + \beta_1^{(i)} x_{ji} + e_{ji}$$

where  $e_{ji}$ ,  $j = 1, \dots, n_i$  are independently and identically distributed  $n(0, \sigma_i^2)$ , for each  $i = 1, \dots, r$ . The classical normal theory approach for testing equal regression models in (6) for each  $i = 1, \dots, r$ ; is to consider the following three nested hypotheses in sequence (Searle 1971, 351):

- (i) without assuming that the regressions for each rater are equal, test the hypothesis

$$(7a) \quad H_0^{(1)} : \sigma_1^2 = \dots = \sigma_r^2 \text{ vs. } H_a^{(1)} : \text{at least one } \sigma_i^2 \neq \sigma_j^2 \text{ for } i \neq j.$$

- (ii) If  $H_0$  in (7a) is accepted, then test

$$(7b) \quad H_0^{(2)}: \beta_1^{(1)} = \dots = \beta_1^{(r)} \text{ vs. } H_a^{(2)}: \text{ at least one } \beta_1^{(i)} \neq \beta_1^{(j)} \text{ for } i \neq j$$

under the assumption that  $\sigma_1^2 = \dots = \sigma_r^2$ .

(iii) If  $H_0$  in (7b) is accepted, then test

$$(7c) \quad H_0^{(3)}: \beta_0^{(1)} = \dots = \beta_0^{(r)} \text{ vs. } H_a^{(3)}: \text{ at least one } \beta_0^{(i)} \neq \beta_0^{(j)} \text{ for } i \neq j$$

under the assumption that  $\sigma_1^2 = \dots = \sigma_r^2$  and  $\beta_1^{(1)} = \dots = \beta_1^{(r)}$ .

If the null hypotheses in (7a), (7b) and (7c) are all accepted then it is concluded that the raters have equal strategies.

These hypotheses are common practice in applications using analysis of covariance. If  $r = 2$ , then the CUSS values can be used to test these hypotheses simultaneously using the test statistic

$$(8) \quad F = n_2 \text{ CUSS}_{n_1} / (n_1 - 2)(1 - \text{CUSS}_{n_1})$$

where

$$\text{CUSS}_{n_1} = \sum_{j=3}^{n_1} r_j^2 / \sum_{j=3}^n r_j^2.$$

Accept  $H_0: \sigma_1^2 = \sigma_2^2, \beta_0^{(1)} = \beta_0^{(2)}$  and  $\beta_1^{(1)} = \beta_1^{(2)}$  at the  $\alpha$  level of significance, if

$$(9) \quad F_{n_1-2, n_2}^{1-\alpha_1} < F < F_{n_1-2, n_2}^{\alpha_2}$$

where  $F_{n_1-2, n_2}^{\beta}$  is the upper  $\beta 100\%$  region of an  $F$  distribution with  $n_1 - 2$

and  $n_2$  degrees of freedom, and  $\alpha_1 + \alpha_2 = \alpha$ . The  $F$  in (8) is only expected to fall into the lower tailed rejection region when  $\sigma_1^2 < \sigma_2^2$ , hence it is suggested that  $\alpha_1 = \alpha/6$  and  $\alpha_2 = 5\alpha/6$ .

If this simultaneous test rejects  $H_0: \sigma_1^2 = \sigma_2^2, \beta_0^{(1)} = \beta_0^{(2)}$  and  $\beta_1^{(1)} = \beta_1^{(2)}$ , then the nested hypotheses (7a), (7b) and (7c) can be used to explore which of these parameters differ significantly.

Let the sequence of scores assessed by raters  $M1, M2, \dots, M_r$  be

$$y_{11}, \dots, y_{n_1 1}, y_{12}, \dots, y_{n_2 2}, \dots, y_{1r}, \dots, y_{n_r r}$$

with corresponding recursive residuals

$$r_1, \dots, r_{n_1}, r_{n_1+1}, \dots, r_{n_1+n_2}, \dots, r_{n_1+\dots+n_{r-1}+1}, \dots, r_{n_1+\dots+n_r}$$

A series of tests using the above recursive residuals and the statistic (9) can be conducted to test the null hypotheses (7a), (7b) and (7c) simultaneously. This hypothesis can be tested at approximately the  $\gamma$  level of significance using the following steps:

**STEP 1:** Test whether  $M1$  and  $M2$  differ in strategy, using (9) and level of significance  $\alpha = \frac{\gamma}{r-1}$ . If  $F$  defined by (8) satisfies (9) then conclude that the raters differ, otherwise assume that raters  $M1$  and  $M2$  adopt the same rating strategy and proceed to **STEP 2**;

STEP j: Define  $\text{CUSS}_{n_1+\dots+n_j} = \sum_{i=3}^{n_1+\dots+n_j} r_i^2 / \sum_{i=3}^{n_1+\dots+n_{j+1}} r_i^2$ . Use

(8) to test whether raters  $M_1, \dots, M_j$  differ in strategy to rater  $M_{j+1}$ . If this  $F$  value satisfies (9) with degrees of freedom  $n_1 + \dots + n_j$  and  $n_{j+1}$ , and  $\alpha = \frac{\gamma}{(r-1)}$  then proceed to STEP  $j+1$ ; otherwise stop and conclude that raters  $M_1$  to  $M_j$  are similar in strategy, but rater  $M_{j+1}$ 's strategy differs significantly from these raters;

⋮

STEP  $r+1$ : Conclude that all raters adopt equivalent strategies at  $\gamma$  level of significance.

It is important to note that these tests can only be justified under the assumptions of model (6). The application in the next section will illustrate that these assumptions may not always be plausible.

## 7. An example of application

A first-year university course with a large enrolment of students is used to illustrate the procedures. The convenor for this course had nine raters at his disposal for the final examination. The examination papers were split up alphabetically amongst raters. The supplementary information was taken as an average of two 2-hour tests written under very similar conditions to the final examination. Although the three-hour examination covered more material than the tests, it was felt that the class mark was a good indicator of a student's ability in the subject, since:

- (a) a detailed marking scheme was provided for each test;

- (b) the students had the tests returned to them together with full solutions and the marking scheme. If students were unhappy about their marks, they returned the examination book, together with an explanation of their grievances. The raters then re-assessed the question(s) and recorded changes;
- (c) the raters were given ample time to complete the assessments.

Any student who had not written both tests was excluded from the analysis.

After the assessments were completed, the sample means for each rater were recorded (table 1).

**TABLE 1: SAMPLE MEANS FOR CLASS MARK AND EXAM MARK.**

| <b>RATER:</b>      | <b>M1</b> | <b>M2</b> | <b>M3</b> | <b>M4</b> | <b>M5</b> | <b>M6</b> | <b>M7</b> | <b>M8</b> | <b>M9</b> |
|--------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| <b>CLASS MARK:</b> | 55,8      | 49,8      | 51,2      | 49,8      | 59,4      | 50,1      | 49,9      | 52,8      | 52,1      |
| <b>EXAM MARK:</b>  | 51,4      | 49,1      | 48,9      | 48,9      | 51,2      | 47,7      | 44,8      | 52,1      | 48,7      |

An analysis of covariance was carried out using the **EXAM MARK** as the response variable ( $y$ ) and the **CLASS MARK** as the covariate. The null hypothesis given by equations (7a) and (7b) were accepted at  $\alpha = 0,05$ . The null hypothesis in (7c) was rejected at the 5% level of significance ( $p = 0,0196$ ). This lead the convenor to the conclusion that the raters' assessments were on average significantly different. Raters 5 and 7 seemed to reflect much lower average **EXAM MARKS** than the average **CLASS MARK**. On this evidence raters 5 and 7 re-assessed some of their examination papers. The average **EXAM MARK** for raters 5 and 7 increased to 54,0 and 45,8 respectively. The null hypothesis (7c) was accepted at  $\alpha = 0,1$  for the revised marks ( $p = 0,3553$ ). The remainder of the analysis was conducted on the revised marks.

### 7.1 Ordinary residuals, and differences $y - x$

The ordinary residuals of **EXAM MARKS** regressed against **CLASS MARKS** (assuming the same regression line for all markers) can be useful. For example, all **EXAM MARKS** corresponding to residuals which are at least two standard deviations from zero can be re-checked for errors. In this application, this amounted to 26 of the 508 examination papers qualifying for re-assessment.

The Shewhart plots for differences between **EXAM MARK** and **CLASS MARK** (i.e.  $y - x$ ) are provided in chronological order of raters in figure 6. Numbers are used to represent the co-ordinates (CUSUM number;  $y - x$ ). These numbers indicate the rater who assessed the respective **EXAM MARKS**. Figure 6 indicates no clear outlying differences. The Box plots of differences  $y_{ji} - x_{ji}$ , for each student  $j$  assessed by rater  $i$ , are recorded in figure 7. It is noticeable that the spread of the scores  $y_{ji} - x_{ji}$  for rater 1 was smaller than the scores for the other raters. One can also detect raters 7 and 5 gave slightly lower scores on average than the other raters. Note that 75% of the differences  $y_{j5} - x_{j5}$  for rater 5 were below zero. The plot of the ordinary residuals against the predicted scores for rater 5 clearly indicates that he/she was more consistent in assessing candidates who score higher than those with lower scores (figure 8a). This also can be seen in figure 8b, the plot of  $y_{ji}$  against  $x_{ji}$ .

The Shewhart plots of the ordinary residuals and the recursive residuals add no new information to figure 6 and are not reported.

### 7.2 Recursive residuals and CUSUM plots

The plot of  $\text{CUSUM}_i$  against  $i$  was obtained using package REGPAC (Galpin 1981). These are slightly different to those discussed in section 4 and hence are interpreted differently. If the recursive residuals are calculated forwards then REGPAC statistical package plots the  $\text{CUSUM}_i$ ;

values in the reverse order. If the recursive residuals are calculated by first reversing the order of the scores (i.e. backwards) then the  $CUSUM_i$  are plotted by reversing the order again (i.e. forwards). The  $CUSUM_i$  plot for the recursive residual calculated backwards shows that for raters M1 to M4 there was very little movement of  $CUSUM_i$  values from the  $CUSUM_i = 0$  line (figure 9). However, note that M5 produced a sharp downward surge. This is an indication that rater M5 assigned lower scores, on average, than other raters. The scores of raters M6 and M8 caused a surge upwards in  $CUSUM_i$  plot. These surges are presumably corrections for M5's and/or M7's scores. The M7 produces a very sharp downward trend indicating that M7 assigns lower scores on average than the other raters. An interesting feature of figure 9 is the fact that sharp surges downwards or upwards of the  $CUSUM_i$  values corresponded to a change in rater.

Figures 10(a)-(i) present, for each rater, the  $CUSUM_i$  plots for the recursive residuals calculated backwards. The interpretation of these plots will be made under the assumption that raters marked the scripts in alphabetical order. The trend of the scores for rater M3 indicates that this rater assigned higher scores on average with each new assessment. Raters M5 and M9 started by assigning higher scores for each new assessment until their scores stabilized and followed a random walk. The upward trend of the scores for raters M1, M2, M4 and M8 indicates that they assigned lower scores on average with each assessment. In general, the behaviour of the rater scores appeared non-stationary, with only two raters showing signs of a stationary assessment strategy, namely raters M6 and M7 (figures 10(f) and (g)). The differences in location (as illustrated by the  $CUSUM_i$  plots against  $i$  in figure 8 and 10) affect the interpretation of the  $CUSAB_i$  plot against  $i$ . In other words, it is difficult to isolate whether the sharp changes in the  $CUSAB_i$  plot are due to differences in spread or differences in re-

gression parameters. For this reason the CUSAB<sub>j</sub> plots are not illustrated.

The fact that the recursive residuals for some raters show non-stationary behaviour suggests that the tests for equality of raters strategy as identified in section 6 may not be justified. Under non-stationarity, the only plausible way of testing equality of strategies would be via some dynamic analysis. This, however will not be covered in this paper. In the next section, the recursive residuals will be assumed stationary to illustrate the tests covered in section 6.

### 7.3 Test of equal raters

Assume that the model relating scores assessed by M1 and M2 to the supplementary information  $x$  is given by equation (6) with  $r = 2$ . The hypothesis that these two raters are equivalent can be tested using statistic (9). The observed value for this test statistic is  $F = 5.57$ , which rejects that raters M1 and M2 have equal rating strategy at  $\alpha = 0.001$ . Similar tests were carried out for the recursive residuals calculated backwards for all pairwise comparisons. The calculation of recursive residuals for comparing raters  $M_i$  with  $M_j$ , was carried out by placing rater  $M_i$ 's sequence of scores first followed by those of  $M_j$ 's if and only if  $i < j$ . Table 2 below summarises the conclusion of these pairwise tests at the  $\alpha = 0.01$  level of significance. Raters M2, M7 and M9 are recorded as significantly different in rater strategy to all other raters.

**TABLE 2: RATERS WHO REFLECT NONSIGNIFICANT DIFFERENCES IN THE PAIRWISE COMPARISON OF RATERS**

|                                                                        |    |    |    |    |    |    |    |    |    |
|------------------------------------------------------------------------|----|----|----|----|----|----|----|----|----|
| RATER                                                                  | M1 | M2 | M3 | M4 | M5 | M6 | M7 | M8 | M9 |
| LIST OF RATERS WHO REFLECT NONSIGNIFICANT DIFFERENCES WITH ABOVE RATER | M3 |    | M1 | M5 | M4 | M3 |    | M4 |    |
|                                                                        |    |    | M6 | M6 | M6 | M4 |    |    |    |
|                                                                        |    |    |    |    | M8 |    | M5 |    |    |

Using the CUSS statistic to test the hypothesis that all raters are equivalent in strategy; it is concluded that raters differ at STEP 1 (see section 6) for  $\gamma = 0,01$ .

Although the conclusions for the tests above are generally the same if the recursive residuals are calculated in the reverse order or if the raters are re-ordered, this cannot be guaranteed. For example, there are four different ways of performing each pairwise test and the conclusions may differ for each. These tests should only be performed for descriptive purposes.

#### 7.4 Concluding remarks

The application in this paper has demonstrated that raters can differ in strategy significantly and that the sequential assessed scores of most raters appear non-stationary. Rater M6's scores showed no non-stationary behaviour. This rater adopted a strategy that assessed all the candidates' performances in question 1 before assessing any responses to question 2, etc. That is, rater M6 reversed the order of assessment from one question to the next. This approach may go some way to alleviating the affects of certain non-stationary strategies. For example, if a rater adopts a strategy that allocates consistently higher marks on average as assessment progresses, then for this rater, each candidate should benefit equally from the strategy of reversal discussed above. These findings suggest that the strategy of assessing all questions of each candidate before assessing any question of the next candidate is ill-advised.

It is worth mentioning that the recursive residuals were not used to analyse the assessments initially. Only the analysis of covariance (ANCOVA) was used. On the basis of the ANCOVA, the assessments were publicised as provisional results. Thereafter, several examination candidates (all assessed by M7) requested a re-evaluation. This revealed that these students should have passed. All candidates failed by M7 were then re-evaluated by the course co-ordinator and the Head of the respective department. As a result of this exercise a further nine students were passed

in the final examination. If the analysis designed in this paper had been carried out in full the questions marked by assessors M7 and M5 would both have qualified for further re-evaluation or adjustment to their assessed scores.

This analysis suggests that the assessment procedure where each rater assesses all questions from a subset of the candidates, should be avoided. Preferably, each rater should assess all candidates' performances in a subset of the questions. Although this does not avoid all problems related to the assessment process, it does seem to be more fair.

If raters become more aware of the subjectiveness of their own assessment strategy, then they can recognize deficiencies in such strategies and hence remediate them. In large assessment tasks, it may be beneficial to:

- (i) select at random (say) ten candidates from those to be assessed;
- (ii) let each rater assess each question of all ten candidates separately;
- (iii) discuss and compare the various strategies (Ballantyne & Sparks 1987).

This would not require too much additional effort and may generate some thought that could speed up later assessments.

Non-stationary behaviour of assessment requires further investigation. In this study such behaviour could not be checked consecutively because it was unknown whether raters assessed candidates in alphabetical order. This behaviour may change for different assessment tasks.

If an examiner wishes to correct for damage already done by raters who changed their strategy sequentially, without re-assessing all candidates with damaged scores, then the examiners must attempt to model the rater's strategy. For example, assume it is known that the strategy of a rater is

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 i + e_i$$

where  $e_1, \dots, e_n$  are iid  $n(0, \sigma^2)$ , then the examiner can adjust each consecutive score ( $y_i$ ) assessed by such a rater, as a result, a fairer assessment of the candidates' performance can be obtained. The adjusted mark for the  $i$ -th candidate and above model is

$$y_i - \beta_2 i.$$

The above model is a special case of Brown *et al.* (1975)'s more general time-trending regressions.

## 8. Discussion

Recursive residuals and CUSUM plots are very useful in sequentially monitoring assessments. This would be particularly useful in large scale assessment tasks such as the marking of provincial matriculation examination scripts in South Africa. The raters of these scripts spend a little under two weeks on the assessment task. These raters generally mark for 'lengthy periods' of time with limited rest. It would be useful to monitor assessment sequentially and determine the maximum length a rater should work before taking a rest. Recursive residuals and CUSUM plots could provide an answer to this. This maximum length may vary from rater to rater and may also depend on the task assigned.

The above procedure could be used in the marking of matriculation examinations in South Africa. In these, seldom is a single rater assigned the task of assessing all scripts for a subsection of the examination. The reason for this is that raters are generally teachers and hence are not allocated to mark scripts from schools at which they are employed. Hence two raters are

generally assigned the task of assessing the same subsections of the scripts. In such situations recursive residuals can be useful in detecting and testing any difference between these raters.

In some provinces in South Africa the year marks for each student are available to the chief examiner for each subject (e.g. Natal Education Department). These can be used as a supplementary variable. If the year marks from the various schools differ in location and also in spread then different recursive residual analyses can be carried out for the schools separately. Past experience such as the previous year's marks of candidates from individual schools could be used to construct recursive residuals. Comparison of raters using recursive residuals may then only be feasible for schools with large numbers of candidates. Assume that two assessors are assigned to mark the same subsections of the examination, then as long as they share the scripts for at least one of these schools, their assessment strategies can be compared using recursive residual techniques. If, for example, the pupils year marks differ in location only, then recursive residuals can be calculated using similar techniques to those in the analysis of covariances. For example, assume there are only two schools and two raters. Let  $y_{ijk}$  be the assessment of the  $i$ -th script from the  $j$ -th school judged by the  $k$ -th rater and let  $x_{ij}$  be the corresponding year mark.

Assume that the first  $\ell$  assessments for rater 1 are all from school 1; that is, the recursive estimated parameters for regression model  $y_{i11} = \beta_{01} + \beta_1 x_{i1} + e_i$  are

$$\hat{\beta}_{01}^{(i)} = \bar{y}_{.11}^{(i)} - \hat{\beta}_1^{(i)} \bar{x}_{.1}^{(i)} \quad ; \quad \bar{y}_{.11}^{(i)} = \sum_{j=1}^i y_{j11} / i ;$$

$$\bar{x}_{.1}^{(i)} = \sum_{j=1}^i x_{j1} / i ; \text{ and}$$

$$\hat{\beta}_1^{(i)} = \frac{\sum_{j=1}^i (x_{j1} - \bar{x}_{.1}^{(i)})(y_{j11} - \bar{y}_{.11}^{(i)})}{\sum_{j=1}^i (x_{j1} - \bar{x}_{.1}^{(i)})^2} \quad i = 3, \dots, \ell.$$

Let the first candidate for school 2, be the  $(\ell + 1)$ -th candidate assessed by rater 1. This observation is used to estimate  $\beta_{02}$  for regression model  $y_{i21} = \beta_{02} + \beta_1 x_{i2} + e_i$  while the estimate of  $\beta_1$  is taken as  $\hat{\beta}_1^{(\ell)}$ , i.e.

$$\hat{\beta}_1^{(\ell+1)} = \hat{\beta}_1^{(\ell)}$$

and

$$\hat{\beta}_1^{(\ell+1)} = y_{121} - \hat{\beta}_1^{(\ell+1)} x_{12}.$$

The recursive residual for this observation is not calculated. Assume, however that the next candidate that was assessed came from school 2, then the recursive residual for this candidate's performance score  $y_{221}$  is given by

$$\frac{y_{221} - \hat{\beta}_{02}^{(\ell+1)} - x_{n_{j+1}j} \hat{\beta}_1^{(\ell+1)}}{\sqrt{1 + \frac{1}{I} + (x_{22} - x_{12})^2 / \sum_{i=1}^{\ell} (x_{i1} - \bar{x}_{.1}^{(\ell)})^2}}$$

Subsequent estimates of regression parameters, after observing  $n_1 (\geq \ell)$  observations from school 1 and  $n_2 (\geq 2)$  observations from school 2, are as follows: (define  $\bar{y}_{.j1}^{(n_j)} = \sum_{\ell=1}^{n_j} y_{\ell j1} / n_j$  and  $\bar{x}_{.j}^{(n_j)} = \sum_{\ell=1}^{n_j} x_{\ell j} / n_j$ )

$$\hat{\beta}_1^{(n_1+n_2)} = \frac{\sum_{j=1}^2 \sum_{\ell=1}^{n_j} (y_{\ell j1} - \bar{y}_{.j1}^{(n_j)})(x_{\ell j} - \bar{x}_{.j}^{(n_j)})}{\sum_{j=1}^2 \sum_{\ell=1}^{n_j} (x_{\ell j} - \bar{x}_{.j}^{(n_j)})^2}$$

$$\hat{\beta}_{0j}^{(n_1+n_2)} = \bar{y}_{.j1}^{(n_j)} - \bar{x}_{.j}^{(n_j)} \hat{\beta}_1^{(n_1+n_2)}$$

for  $j = 1, 2$ . Assume that the  $(n_1 + n_2 + 1)$ -th candidate comes from school  $j$  then the recursive residual for his/her assessed score  $y_{n_j+1j1}$  is

$$\frac{y_{n_j+1j1} - \hat{\beta}_{0j}^{(n_1+n_2)} - x_{n_j+1j} \hat{\beta}_1^{(n_1+n_2)}}{\sqrt{1 + \frac{1}{n_j} + (x_{n_j+1j} - \bar{x}_{\cdot j}^{(n_j)})^2 / \sum_{j=1}^2 \sum_{i=1}^{n_j} (x_{ij} - \bar{x}_{\cdot j}^{(n_j)})^2}}$$

for  $j = 1, 2$ .

On the whole, the analysis of assessment strategies of raters is a neglected area of educational research and deserves more attention.

## Acknowledgement

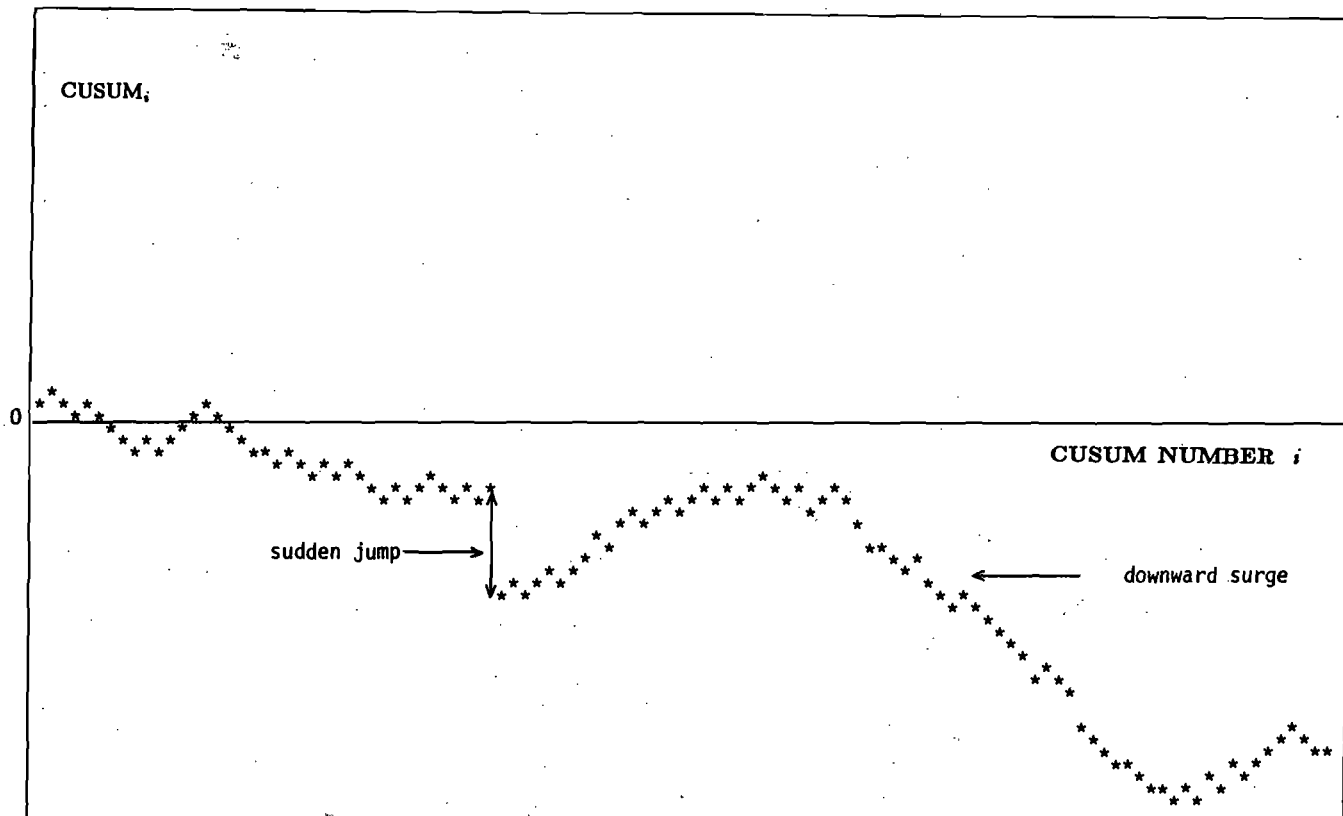
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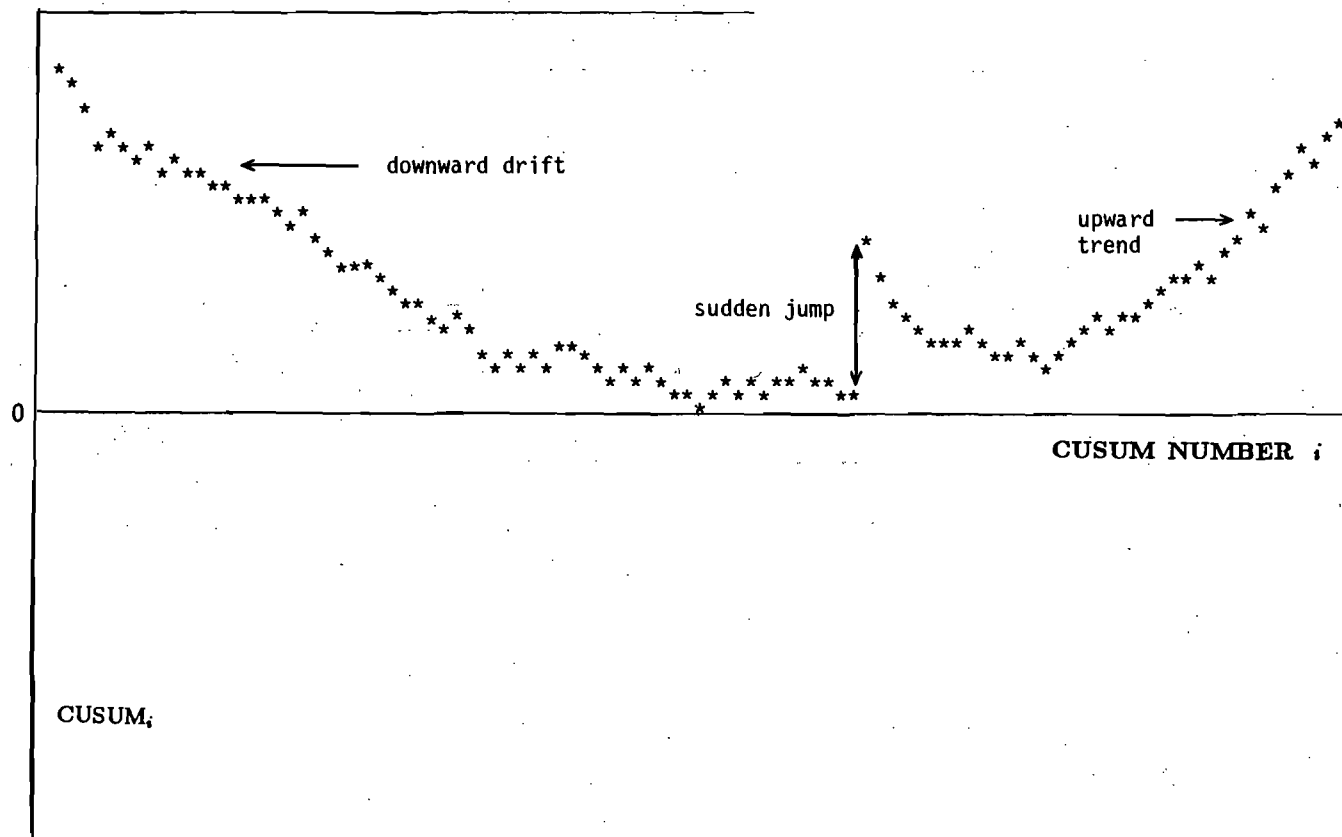
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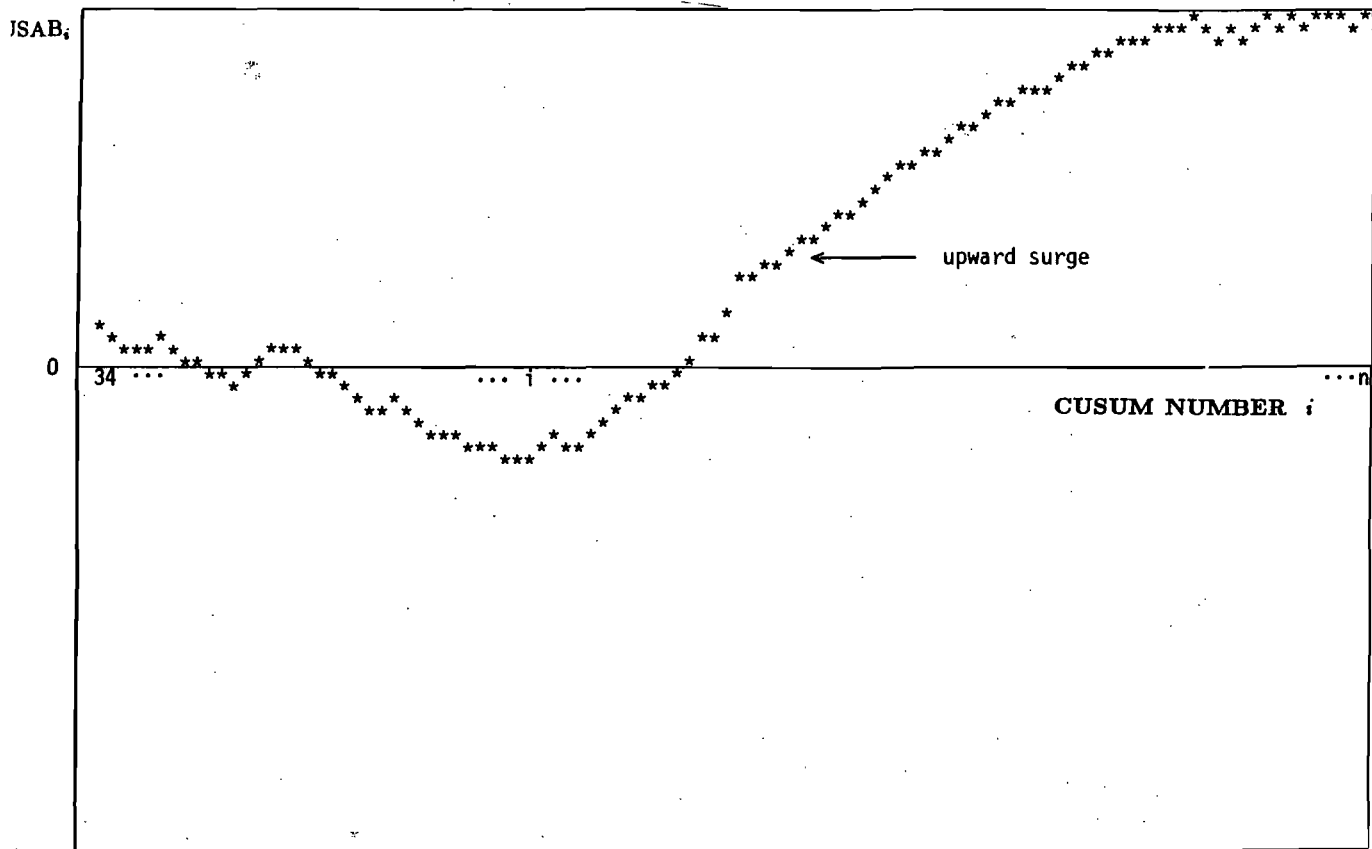
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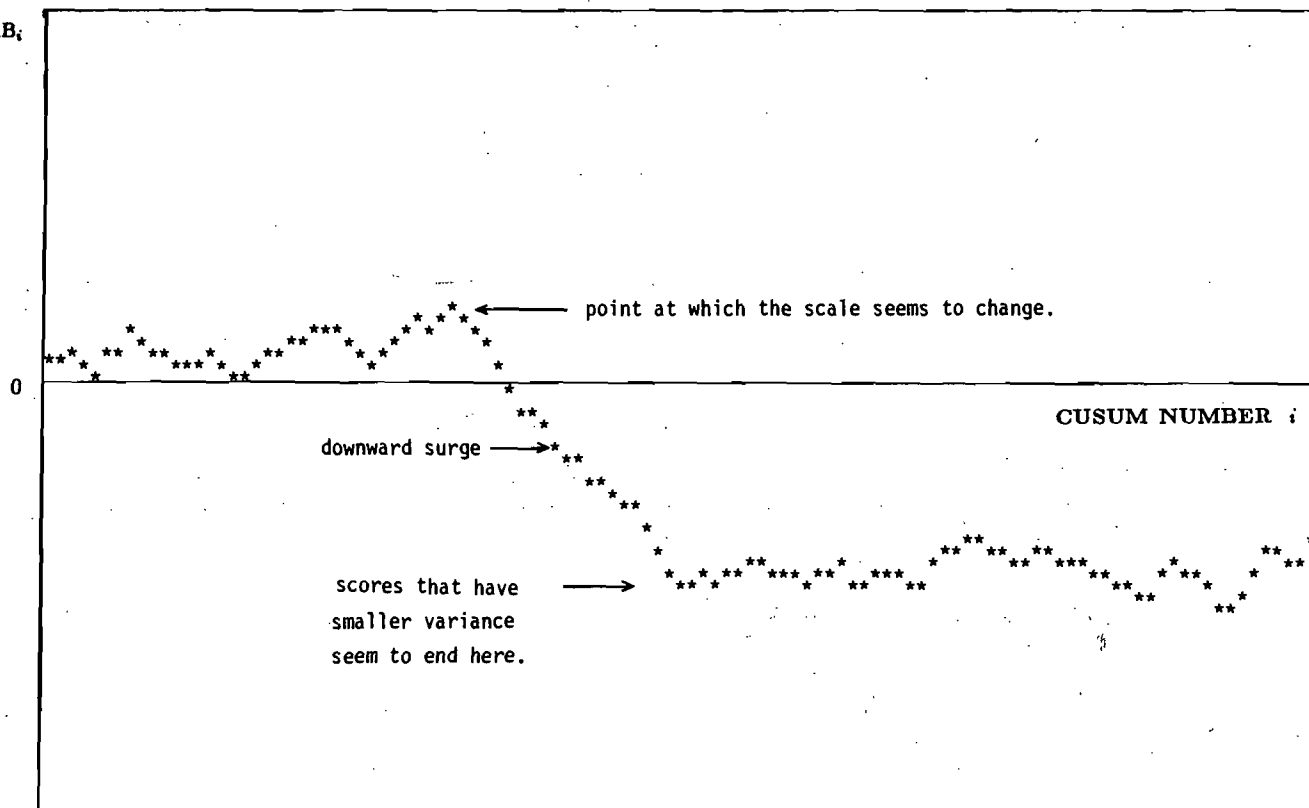
**FIGURE 1: CUSUM<sub>i</sub> PLOT AGAINST  $i$**



**FIGURE 2:  $CUSUM_i$  PLOT AGAINST  $i$**



**FIGURE 3: CUSAB<sub>i</sub> PLOT AGAINST i**

SAB<sub>i</sub>

**FIGURE 4: CUSAB<sub>i</sub> PLOT AGAINST i**

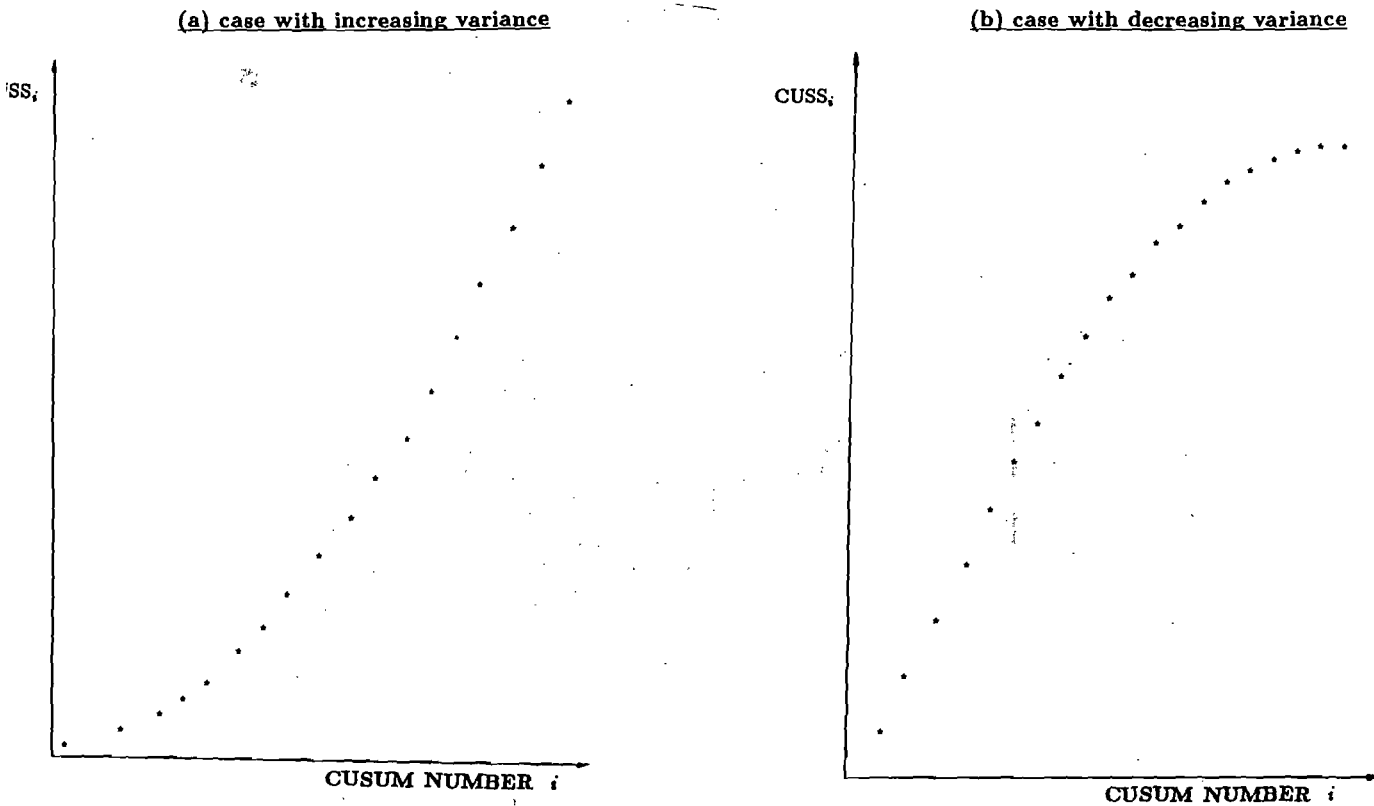
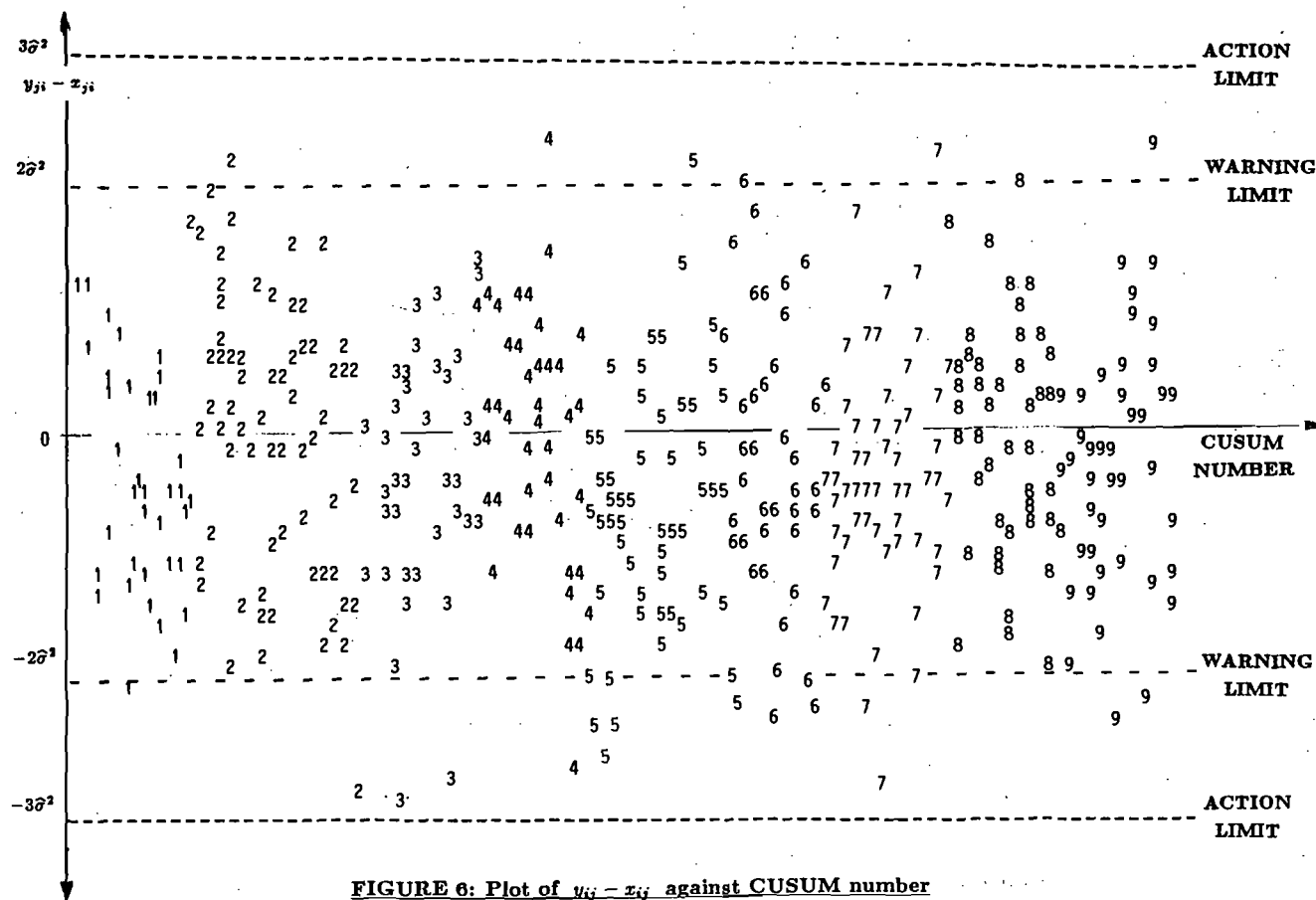


FIGURE 5: CUSS <sub>$i$</sub>  PLOT AGAINST  $i$



**FIGURE 6: Plot of  $v_{ji} - x_{ji}$  against CUSUM number**  
for  $i = 1, \dots, 9$  &  $j = 1, \dots, n_i$

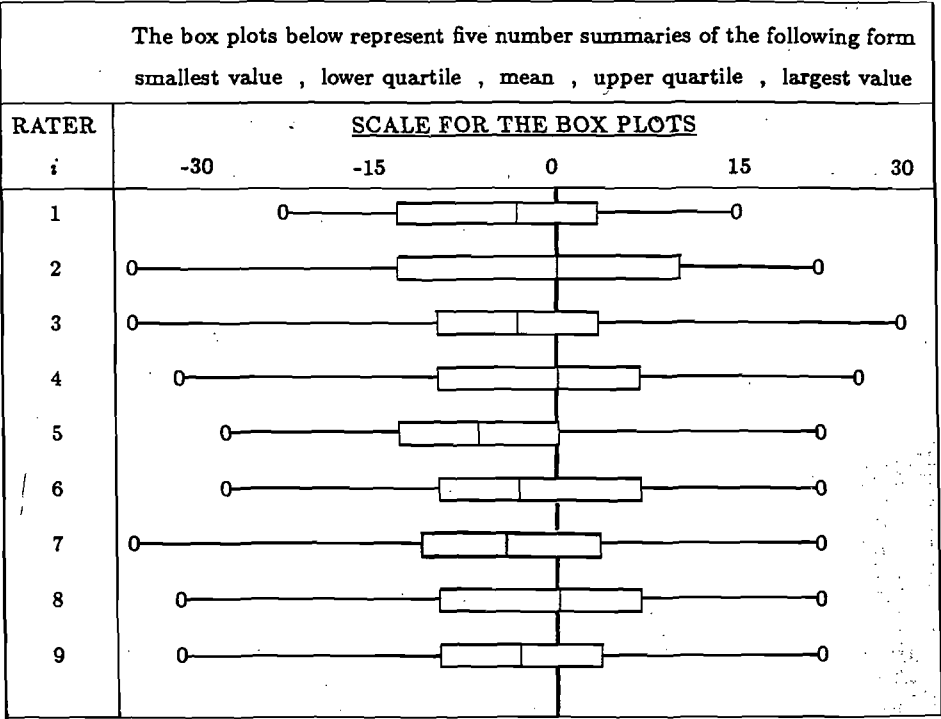


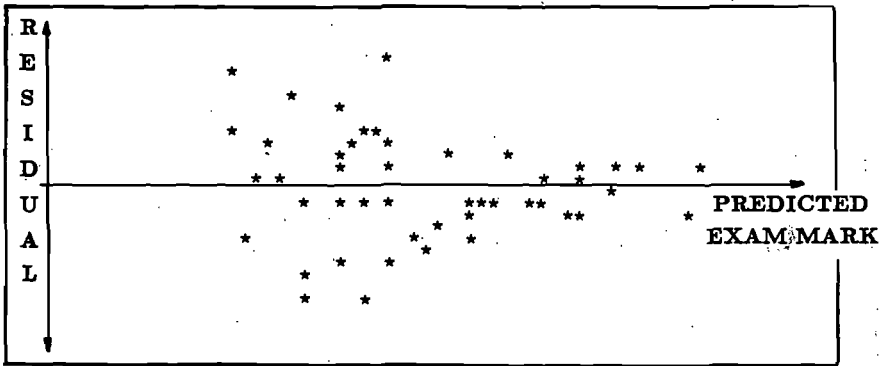
FIGURE 7

The box plots are used to describe the empirical distribution of the difference:

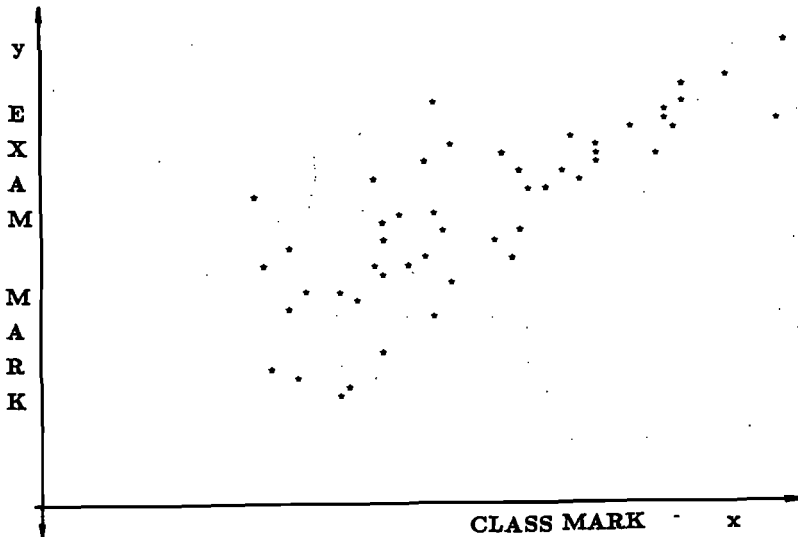
$y_{ji} - x_{ji} ; j = 1; \dots; n_i$

for each rater separately i.e. for each  $i = 1, \dots, 9$

(a) Residual vs Predicted EXAM MARK



(b) EXAM MARK vs CLASS MARK



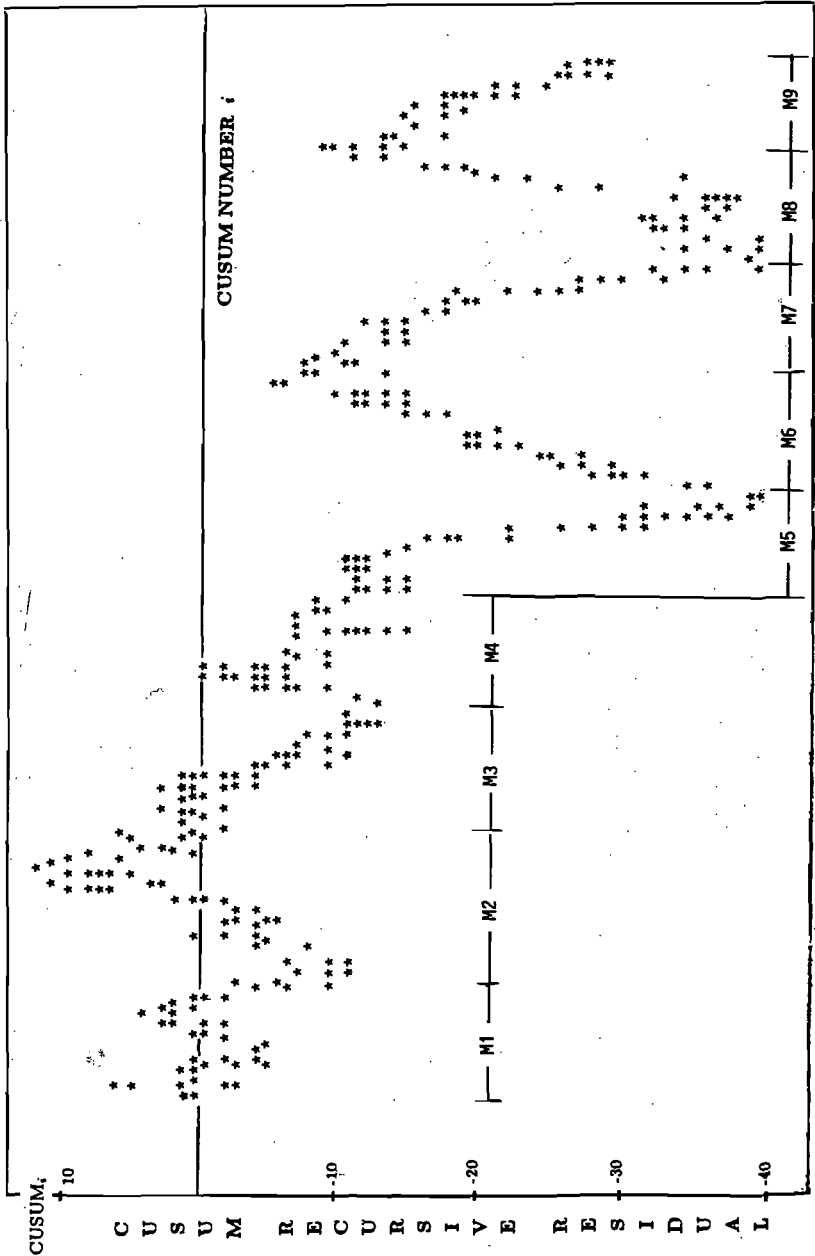


FIGURE 9  
Plot CUSUM RESIDUAL/SD against CUSUM NUMBER  
recursive residuals calculated backwards

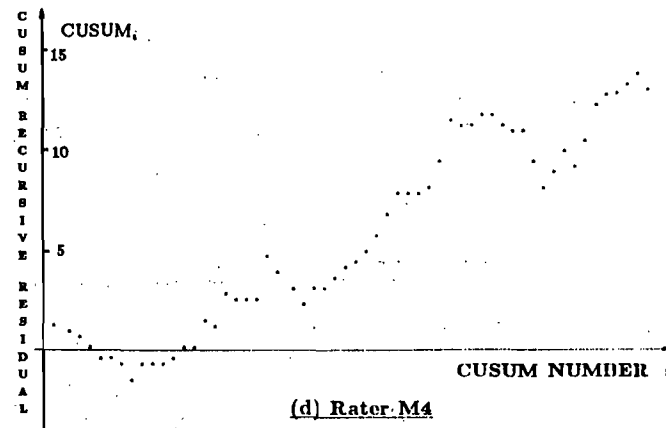
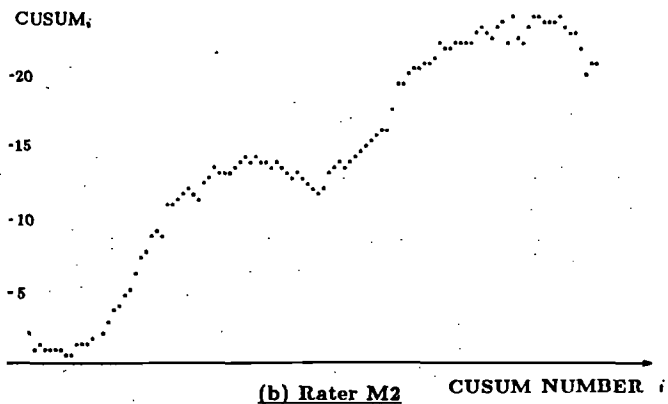
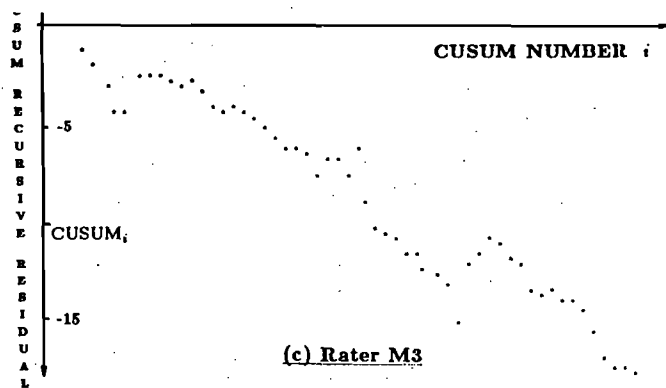
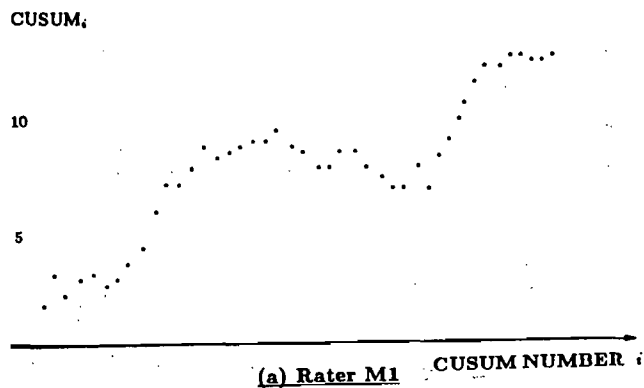
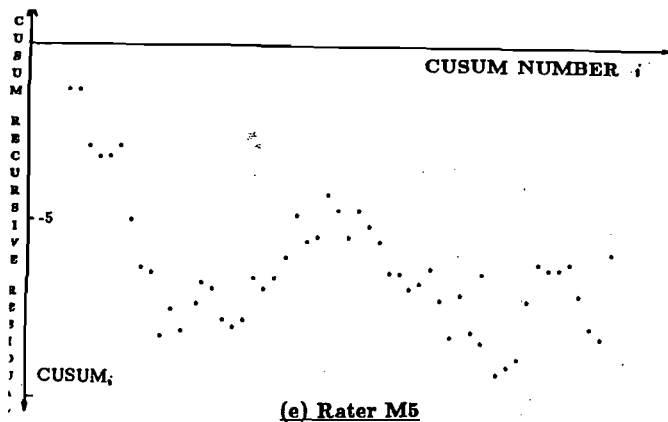
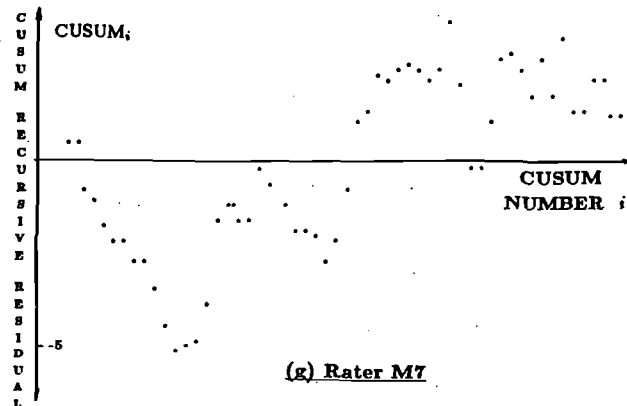


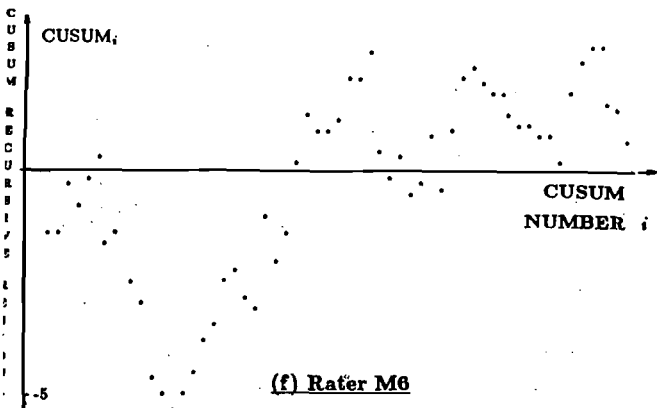
FIGURE 10 (a), (b), (c) and (d). CUSUM<sub>i</sub> PLOT AGAINST  $i$  recursive residuals calculated backwards



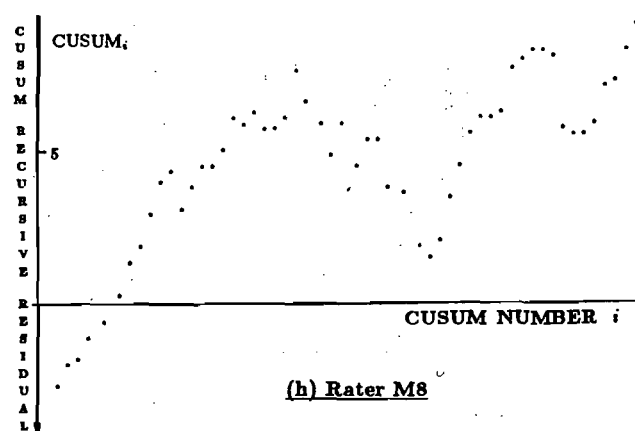
(e) Rater M5



(g) Rater M7

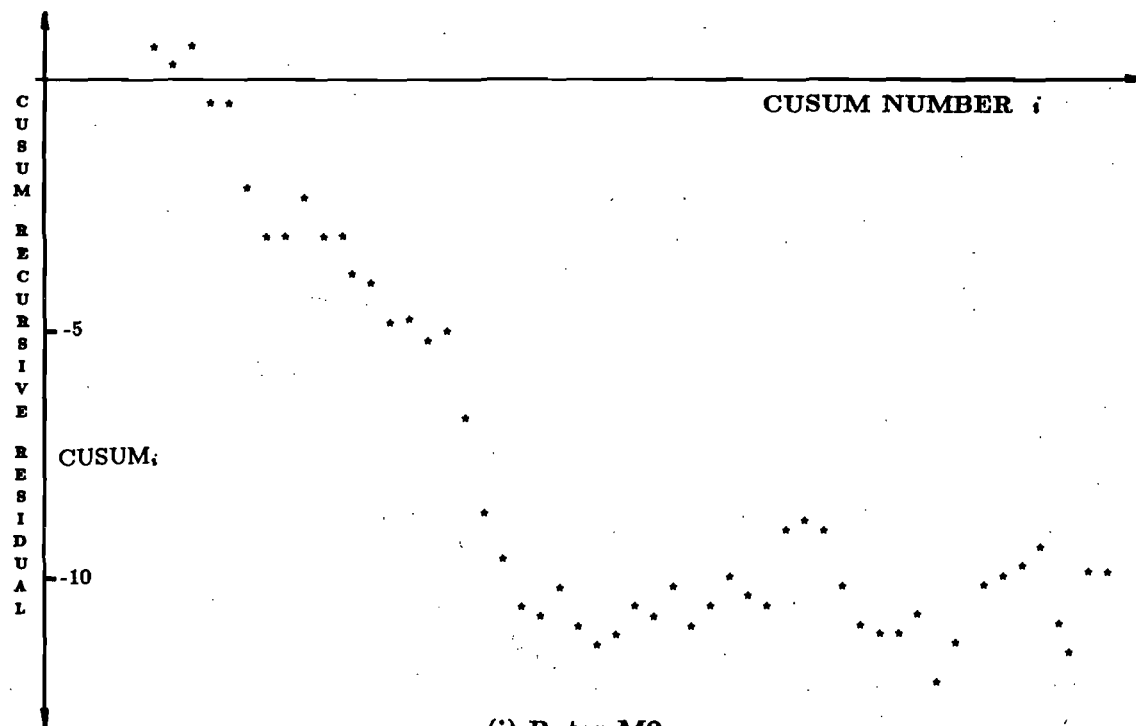


(f) Rater M6



(h) Rater M8

FIGURE 10 (e), (f), (g) and (h).  $CUSUM_i$  PLOT AGAINST  $i$  recursive residuals calculated backwards



(i) Rater M9

FIGURE 10 (i).  $CUSUM_i$  PLOT AGAINST  $i$  recursive residuals calculated backwards